Prediction for New Patients

Fold-in Update for the Patient Loadings $l_{\star k}$

Supplement to *Supervised Bayesian Matrix Factorization: Corrected Update Derivations* (March 2026)

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1 Introduction and Setting

The training algorithm of the March document returns posterior moments for L, F, $\boldsymbol{\beta}$ and the precisions τ. The loadings l_{ik} exist only for patients in the training set, so a patient who was not used for fitting cannot be scored from them. This note derives the update that infers the loadings $l_{\star k}$ of a new patient $\star$ by re-using the L-update of Section 5 with everything else frozen ("fold-in"), and gives the resulting predictor $\eta_\star = \sum_k l_{\star k}\beta_k$. The notation follows Section 2.2: a bar denotes a posterior mean and an overline a posterior second moment, e.g. $\overline{f_{jk}^2} = \mathrm{Var}_q(f_{jk}) + \bar{f}_{jk}^2$.

Frozen training output. After the training algorithm has converged, we keep

$$\bar{f}_{jk},\ \overline{f_{jk}^2}\ (j = 1, \dots, p), \qquad \tau_j, \qquad \hat{g}_{l_k}, \qquad \bar{\beta}_k,\ \overline{\beta_k^2} \qquad (k = 1, \dots, K),$$

where $\hat{g}_{l_k}$ is the fitted prior on the loadings of factor k. The new patient has genomics data $Y_{\star j}$ for $j \in \mathcal{G} \subseteq \{1, \dots, p\}$, standardized in the same way as the training data (within its own cohort); genes outside $\mathcal{G}$ are simply absent from every sum below.

2 Model and Objective for the New Patient

The genomics model of Eq. (3) applied to patient $\star$ is

$$Y_{\star j} = \sum_{k=1}^{K} l_{\star k}\, f_{jk} + \varepsilon_{\star j}, \qquad \varepsilon_{\star j} \sim \mathcal{N}\left(0, \tau_j^{-1}\right), \qquad l_{\star k} \overset{\text{iid}}{\sim} \hat{g}_{l_k}, \tag{1}$$

and the survival model of Eq. (4) gives the linear predictor

$$\eta_\star = \boldsymbol{l}_\star^\top \boldsymbol{\beta} = \sum_{k=1}^{K} l_{\star k}\beta_k. \tag{2}$$

With q_f, τ and $\hat{g}_{l_k}$ held at their training values, the objective for $q(l_{\star k})$ is

$$\mathcal{F}(q_{l_\star}) = \mathbb{E}_{q_l, q_f}[\log P(Y_\star \mid \boldsymbol{l}_\star, F, \tau)] + \sum_k \mathbb{E}_{q_{l_{\star k}}}\left[\log \frac{\hat{g}_{l_k}(l_{\star k})}{q_{l_{\star k}}(l_{\star k})}\right]. \tag{3}$$

There is no survival term: the outcome $(t_\star, \delta_\star)$ of a new patient is not available.

Revision Note

[P1] Training and prediction are not the same map under the March update. The training update for l_{ik} (Eqs. (39)–(40)) contains the survival contributions $W_{ii}\overline{\beta_k^2}$ and $W_{ii}z_i^{-k}\bar{\beta}_k$. At prediction these are absent ($A_{\star k}^{\text{surv}} = B_{\star k}^{\text{surv}} = 0$), so the training loadings $\bar{l}_{ik}$ borrow information from the patient's own outcome while $\bar{l}_{\star k}$ does not. This can make the in-sample fit of $\boldsymbol{\beta}$ optimistic, by an amount that depends on the size of the survival terms, and it creates a train/test discrepancy. *Recommended fix:* after convergence, recompute genomics-only loadings for every training patient with the procedure of Sections 5 and 7 (F, τ, $\hat{g}_{l_k}$ fixed, only Y_i used), and refit $\boldsymbol{\beta}$ by Eqs. (80)–(84) using the resulting $\bar{l}_{ik}$ and $\overline{l_{ik}^2}$. If $W_{ii}\overline{\beta_k^2} \ll \sum_j \tau_j \overline{f_{jk}^2}$ the two versions coincide to numerical precision, which can be checked directly.

3 Genomics Likelihood Contribution

Define the partial residual of patient $\star$ exactly as in Eqs. (27)–(28), using the current posterior means of the other loadings:

$$R_{\star j}^{-k} = Y_{\star j} - \sum_{k' \neq k} \bar{l}_{\star k'}\, \bar{f}_{jk'}, \qquad \Rightarrow \qquad Y_{\star j} \approx R_{\star j}^{-k} + l_{\star k} f_{jk}. \tag{4}$$

Derivation Steps

The log-likelihood of patient $\star$, feature j is

$$\log P(Y_{\star j} \mid \boldsymbol{l}_\star, f_{j\cdot}, \tau_j) = -\tfrac{1}{2}\tau_j \Big(Y_{\star j} - \sum_k l_{\star k} f_{jk}\Big)^2 + C. \tag{5}$$

Substituting $Y_{\star j} = R_{\star j}^{-k} + l_{\star k} f_{jk}$ and expanding the square,

$$\log P(Y_\star \mid \cdot) = \sum_{j \in \mathcal{G}} \Big[-\tfrac{1}{2}\tau_j \Big(\big(R_{\star j}^{-k}\big)^2 - 2 R_{\star j}^{-k}\, l_{\star k} f_{jk} + l_{\star k}^2 f_{jk}^2 \Big)\Big]. \tag{6}$$

The term $(R_{\star j}^{-k})^2$ is constant with respect to $l_{\star k}$ and is absorbed into C. Taking the expectation $\mathbb{E}_{q_f}[\cdot]$ over the posterior of F (and using $\mathbb{E}_{q_f}[f_{jk}^2] = \overline{f_{jk}^2}$ as in Revision Note R3):

$$\longrightarrow \sum_{j \in \mathcal{G}} \Big[\tau_j R_{\star j}^{-k}\, l_{\star k} \underbrace{\mathbb{E}_{q_f}[f_{jk}]}_{\bar{f}_{jk}} - \tfrac{1}{2}\tau_j\, l_{\star k}^2 \underbrace{\mathbb{E}_{q_f}[f_{jk}^2]}_{\overline{f_{jk}^2}} \Big]. \tag{7}$$

Collecting the coefficients of $l_{\star k}$ (term B) and $l_{\star k}^2$ (term A):

$$A_{\star k}^{\text{gen}} = \sum_{j \in \mathcal{G}} \tau_j\, \overline{f_{jk}^2}, \qquad B_{\star k}^{\text{gen}} = \sum_{j \in \mathcal{G}} \tau_j\, R_{\star j}^{-k}\, \bar{f}_{jk}. \tag{8}$$

Since there is no survival term for the new patient, the combined coefficients are the genomics coefficients alone:

$$\boxed{A_{\star k} = \sum_{j \in \mathcal{G}} \tau_j\, \overline{f_{jk}^2}} \qquad \boxed{B_{\star k} = \sum_{j \in \mathcal{G}} \tau_j\, R_{\star j}^{-k}\, \bar{f}_{jk}} \tag{9}$$

Note that $A_{\star k}$ depends on the patient only through the set of measured genes $\mathcal{G}$, so all patients of a cohort that measures the same genes share the same $A_{\star k}$. A gene that is not measured removes its term from both sums, which lowers $A_{\star k}$, enlarges the standard error below, and pulls the posterior towards the prior.

4 EBNM Problem with a Fixed Prior

The quadratic form $-\frac{1}{2}A_{\star k}l_{\star k}^2 + B_{\star k}l_{\star k}$ is equivalent to a Gaussian likelihood $l_{\star k} \mid x_{\star k}, s_{\star k} \sim \mathcal{N}(x_{\star k}, s_{\star k}^2)$ with pseudo-observation $x_{\star k} = B_{\star k}/A_{\star k}$ and pseudo-variance $s_{\star k}^2 = 1/A_{\star k}$, exactly as in Section 5.5.

EBNM Problem

For each factor k, the new-patient update solves the normal-means problem with the prior *fixed* at its training value:

$$x_{\star k} = \frac{B_{\star k}}{A_{\star k}} = \frac{\sum_{j\in\mathcal{G}} \tau_j R_{\star j}^{-k} \bar{f}_{jk}}{\sum_{j\in\mathcal{G}} \tau_j \overline{f_{jk}^2}}, \qquad s_{\star k} = \frac{1}{\sqrt{A_{\star k}}}, \tag{10}$$

$$\hat{q}_{l_{\star k}}(l) \;\propto\; \mathcal{N}\big(x_{\star k};\, l,\, s_{\star k}^2\big)\, \hat{g}_{l_k}(l), \qquad \Big(\bar{l}_{\star k},\, \overline{l_{\star k}^2}\Big) = \mathrm{EBNM}_{\hat{g}_{l_k}}(x_{\star k}, s_{\star k}). \tag{11}$$

The solver returns the posterior moments $\bar{l}_{\star k} = \mathbb{E}_{\hat{q}}[l_{\star k}]$ and $\overline{l_{\star k}^2} = \mathrm{Var}_{\hat{q}}(l_{\star k}) + \bar{l}_{\star k}^2$.

Revision Note

[P2] The prior is not re-estimated. In training, EBNM$(\boldsymbol{x}, \boldsymbol{s})$ receives n pseudo-observations per factor and returns both $\hat{g}_{l_k}$ and $\hat{q}_{l_k}$ (Eq. (43)). A single new patient provides one pseudo-observation per factor, which cannot identify a prior, and refitting $\hat{g}_{l_k}$ would also change the model that $\boldsymbol{\beta}$ was estimated under. The new-patient update therefore returns only the posterior under the fixed $\hat{g}_{l_k}$.

Remark: Point-Exponential Prior

If $\hat{g}_{l_k} = (1-\pi)\delta_0 + \pi\,\lambda e^{-\lambda l}\mathbf{1}\{l > 0\}$ (a nonnegative prior), then with $m = x_{\star k} - \lambda s_{\star k}^2$, $a = m/s_{\star k}$ and $r = \phi(a)/\Phi(a)$,

$$p_{\hat{g}}(x_{\star k}) = (1-\pi)\mathcal{N}(x_{\star k}; 0, s_{\star k}^2) + \pi\lambda\, e^{-\lambda x_{\star k} + \lambda^2 s_{\star k}^2/2}\, \Phi(a), \tag{12}$$

$$\omega_+ = 1 - \frac{(1-\pi)\mathcal{N}(x_{\star k}; 0, s_{\star k}^2)}{p_{\hat{g}}(x_{\star k})}, \tag{13}$$

$$\bar{l}_{\star k} = \omega_+\,(m + s_{\star k} r), \qquad \overline{l_{\star k}^2} = \omega_+\,\big(m^2 + s_{\star k}^2 + m\, s_{\star k} r\big). \tag{14}$$

The posterior is $\omega_0\delta_0$ plus ω_+ times $\mathcal{N}(m, s_{\star k}^2)$ truncated to $(0, \infty)$, with $\omega_0 = 1 - \omega_+$.

5 Coordinate Ascent over the Factors

The K loadings of the new patient are coupled through the shared residual in Eq. (4), so the update is cycled over $k = 1, \ldots, K$ until the posterior means stop changing. Each step maximizes $\mathcal{F}(q_{l_\star})$ exactly over $q_{l_{\star k}}$ with the other factors fixed, so $\mathcal{F}$ is non-decreasing across steps. The stopping rule is

$$\Delta_l = \max_k \left| \bar{l}_{\star k}^{\text{new}} - \bar{l}_{\star k}^{\text{old}} \right| < \text{tol}. \tag{15}$$

Initialize at $\bar{l}_{\star k} = 0$ (or at the nonnegative least-squares fit); a handful of sweeps is typically enough.

Batch form. For n_{new} patients from one cohort with common $\mathcal{G}$, $A_{\star k}$ is a scalar shared by all patients and $B_{\cdot k} = \left(R^{-k}\,\text{diag}(\tau_{\mathcal{G}})\right)\bar{\boldsymbol{f}}_{\mathcal{G},k}$ is a vector of length n_{new}, so one EBNM call per factor and per sweep handles the whole batch with the prior held fixed.

6 Predictor and its Uncertainty

With the converged moments, the posterior mean of the linear predictor is

$$\boxed{\bar{\eta}_\star = \sum_{k=1}^{K} \bar{l}_{\star k}\,\bar{\beta}_k} \tag{16}$$

and, because q factorizes over k and $l \perp \beta$ under q,

$$\text{Var}_q(\eta_\star) = \sum_{k=1}^{K} \left[\overline{l_{\star k}^2}\;\overline{\beta_k^2} - \bar{l}_{\star k}^2\,\bar{\beta}_k^2 \right] = \sum_k \left[\overline{\beta_k^2}\,\text{Var}_q(l_{\star k}) + \bar{l}_{\star k}^2\,\text{Var}_q(\beta_k) \right]. \tag{17}$$

This is the new-patient counterpart of the error-in-variables term $\text{Var}_q(l_{ik})\beta_k^2$ in Eq. (69). Higher $\eta_\star$ denotes higher hazard (Eq. (4)). Because the Cox model depends on η only through differences between patients, a common shift of the scores (for example from centering) does not change the concordance index.

7 Complete Algorithm Summary

Prediction for a New Patient — Fold-in Algorithm

Inputs (frozen from training): $\bar{f}_{jk}$, $\overline{f_{jk}^2}$, τ_j, $\hat{g}_{l_k}$, $\bar{\beta}_k$, $\overline{\beta_k^2}$; data $Y_{\star j}$, $j \in \mathcal{G}$, standardized within the patient's cohort; tolerance tol.
Initialize: $\bar{l}_{\star k} = 0$ for $k = 1, \ldots, K$.
Repeat until $\Delta_l <$ tol:

1. For $k = 1, \ldots, K$:
   (a) $R_{\star j}^{-k} = Y_{\star j} - \sum_{k' \neq k} \bar{l}_{\star k'} \bar{f}_{jk'}$ for $j \in \mathcal{G}$.
   (b) $A_{\star k} = \sum_{j \in \mathcal{G}} \tau_j \overline{f_{jk}^2}, \quad B_{\star k} = \sum_{j \in \mathcal{G}} \tau_j R_{\star j}^{-k} \bar{f}_{jk}$.
   (c) $\left(\bar{l}_{\star k}, \overline{l_{\star k}^2}\right) \leftarrow \mathrm{EBNM}_{\hat{g}_{l_k}}\big(x = B_{\star k}/A_{\star k},\ s = 1/\sqrt{A_{\star k}}\big)$.
2. $\Delta_l = \max_k \left|\bar{l}_{\star k}^{\mathrm{new}} - \bar{l}_{\star k}^{\mathrm{old}}\right|$.

Output: $\bar{\eta}_\star = \sum_k \bar{l}_{\star k} \bar{\beta}_k$ and $\mathrm{Var}_q(\eta_\star) = \sum_k \left[\overline{l_{\star k}^2}\, \overline{\beta_k^2} - \bar{l}_{\star k}^2 \bar{\beta}_k^2\right]$.